

July 5, 2026

Editor
The R Journal

Dear Emi Tanaka,

Please consider our article titled “`svmodt`: An R Package for Linear SVM-Based Oblique Decision Trees” for publication in the R Journal.

The manuscript introduces `svmodt`, a new package implementing oblique decision trees that use linear Support Vector Machines at each node to determine multivariate splits. As part of this work we also provide `STreeR`, the first fully native R implementation of the `STree` algorithm, and validate it against the original Python implementation. `svmodt` extends this foundation with dynamic feature selection strategies, node-specific class weighting for imbalanced data, and feature-diversity mechanisms through penalisation. We show that `svmodt` outperforms axis-parallel decision trees on benchmark datasets while producing trees of comparable structure and interpretability.

We believe readers of the R Journal will find this article useful, as oblique decision trees remain underrepresented among native R implementations, and `svmodt` offers both a validated `STree` port and meaningful methodological extensions with full documentation and reproducible examples.

Regards,

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RjTools' Automatic Check Results

SUCCESS: 13 | NOTE: 2 | WARNING: 0 | ERROR: 0

- NOTE: Remove `.log`, `.aux` and `.out` files before submitting.
- NOTE: A potential list of spelling to check: interpretability, `svmodt`, penalisation, interpretable, hyperplanes, multiclass, `scikit`, `learn`'s, vectorised, generalises, prioritised, randomised, hypercubes, generalisation, optimisation, perceptron, predominatly, maximised, regularisation, quadratically, recurses, scalability, labelled, benchmarking, accuracies, `svm`, pre, behaviour, characterised, stochasticity, `tourr`, visualisation, coloured, underperforms, utilise, penalised, penalising, unpenalised, localised, underfits, underfitting, centre, auditable, visualises, comparators, optimise, favour, prioritise, adaptively, `rpart`, untuned, overfit, `mtry`, prioritises, scalable, parallelised, organisation, `wdbc`, echocardiogram, `ctg`